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SUMMARY

Full-stack engineer, 2 years shipping production systems in government, fintech, and healthcare — from OCR pipelines processing government archives to payment infrastructure handling real transaction volume. Comfortable owning a feature end-to-end: schema, API, deployment, on-call.

SKILLS

- **Languages:** C#, TypeScript, JavaScript, Go, Python, C++, SQL
- **Frameworks & Libraries:** React.js, React Native, ASP.NET Core, Entity Framework Core, OpenCV, EasyOCR, PaddleOCR
- **Architecture & APIs:** REST, Microservices, Clean Architecture, JWT, idempotency patterns
- **Databases & Cloud:** SQL Server, Redis, MongoDB, GCP (Compute Engine, Cloud Run, Storage)
- **DevOps & Observability:** Docker, GitHub Actions, GitLab CI, Linux, Grafana, Prometheus

EDUCATION

- **Zewail City University of Science and Technology** Sep 2023 – Expected June 2027
B.Sc., Computational Sciences and Artificial Intelligence
Relevant Coursework: Data Structures, Algorithms, Object Oriented Programming, Operating Systems, Databases, Computer Networks

ACHIEVEMENTS

- 4th place, university round, ECPC (Egyptian Collegiate Programming Contest)

EXPERIENCE

- **Ministry of Endowments** | Full-Stack Developer Jun 2025 – Present
Cairo, Egypt, concurrent with university
Government ministry managing a national portfolio of 100,000+ real estate and agricultural endowments.
 - **Sukuk Donations Platform** | ASP.NET Core, React, SQL Server
 - * Architected a sukuk (Islamic bond) donations platform covering CRM, ERP, inventory, and distribution across multiple endowment units; live in production, generating over 100M EGP in profit to date
 - * Designed the platform's data model to keep donor management, endowment unit accounting, physical asset inventory, and disbursement workflows consistent with each other while respecting Islamic finance compliance rules around fund segregation
 - **Handwritten Arabic OCR Pipeline** | Python, OpenCV, EasyOCR, PaddleOCR
 - * Engineered an OCR pipeline for scanned handwritten Arabic ownership documents, from image preprocessing (OpenCV) through to a structured DOCX output per document
 - * Tuned preprocessing and OCR ensemble to reach 99% accuracy per document, with particular attention to Arabic ligatures, historically the hardest failure mode for handwritten Arabic OCR
 - * Cut a batch processing job from five hours to under ten minutes
 - **Live TV Competition App (Dawry Al-A'imma Al-Nujaba)** | React, ASP.NET Core
 - * Owned the frontend end-to-end for a live TV competition app, running real-time answers and scoring across multiple touchscreens synced via webhooks
 - * Aired live on national television across two events to date, with strong audience feedback
 - **Employees Transfer Management System** | React, SQL Server, Python, OpenCV, EasyOCR
 - * Developed the frontend for a yearly system managing fair and transparent employee transfers between government directorates nationwide, moving thousands of employees per cycle
 - * Implemented flows for HR validation, automated scoring, managerial approvals, and administrative auditing against a secure relational database
 - * Added an embedded OCR module verifying submitted documents for validity at over 95% correctness, using OpenCV preprocessing, EasyOCR for Arabic, and fuzzy matching to handle inconsistent document formatting

Events Management Platform | *React, ASP.NET Core*

- * Delivered the frontend for a platform hosting real-world events with students attending from around the world, including automated email notifications and on-site attendance verification via student location
- * Took full ownership of the Zoom integration, backend and frontend, letting VIPs join and participate remotely regardless of location

Parliament Correspondence System | *React, ASP.NET Core, SQL Server*

- * Implemented a system for communication between parliament members and the ministry, handling archival and automatic reply routing of correspondence
- * Designed a Word-like WYSIWYG editor modal with templated responses, making it easy for non-technical staff to draft and send replies

Hierarchy Management SPA | *React, ASP.NET Core*

- * Created a hierarchy management SPA for rendering and coordinating large org trees across ministry departments
- * Handled deeply nested, department-specific org structures with permission-scoped views, so different ministry roles only saw and edited the branches relevant to them, and kept the tree responsive despite its size
- Shipped a mosque outbuilding lease management system (React, ASP.NET Core, SQL Server), validating contract end dates and automatically notifying stakeholders to streamline a previously manual leasing process
- Core contributor to a shared design system and component library (React, React Native) used across multiple government web and mobile products
- Worked directly with non-technical ministry stakeholders across departments to turn policy and procedural requirements into concrete data models and workflows
- **Innova Platform** | Founding Engineer (Back-End & DevOps) Aug 2025 – Aug 2026
Remote, concurrent with university
 - Joined as a founding engineer; contributed to core backend architecture decisions using ASP.NET Core (clean architecture), including data modeling, auth, and API contracts
 - Built a modular data layer on Entity Framework Core and Redis, keeping average API latency under 200ms as feature surface grew over six months
 - Implemented secure payment flows with AES encryption and idempotency keys to prevent duplicate charges on retried requests
 - Owned CI/CD pipeline (GitHub Actions), Dockerized services, and Linux VPS operations end-to-end
- **Freelance** | Full-Stack Developer Jan 2023 – Present
Remote

Aqua Group | *React, React Native*

- * Developed internal systems for aquaculture and fish farm management, including a React web dashboard and a companion React Native mobile app

Retail Inventory & POS System | *React, ASP.NET Core, SQL Server*

- * Designed an offline-capable retail inventory and POS-linked stock control system that noticeably cut manual stock discrepancies

Clinic Operations Platform | *React, ASP.NET Core, SQL Server*

- * Delivered full-stack clinic operations platforms for web and desktop, with REST backends and SQL Server

PROJECTS

- **Clinical Drug Interaction System** | ASP.NET Core, React, SQL Server, EF Core
 - Designed a clinical decision engine covering 10k+ drug combinations; chose a graph traversal approach over naive joins to achieve sub-200ms response times at scale
 - Built ETL pipeline from FDA data sources into SQL Server; delivered React dashboard with advanced filtering for use in live clinical safety workflows